```
    elif lgtper >= 10.00 and lgtper < 80.00:
        bledon(rled,gled,bled)
    else:
        gledon(rled,gled,bled)

    sleep(1)
  except:
    aledoff(rled,gled,bled)
    machine.reset()
```

Running a web server with PI Pico

As mentioned earlier, Pico 1/2 series boards are available with wireless capabilities too. We'll explore Pico W networking with simple programs to get you started and apply the same concepts to Pico 2W as well. We'll use the board as an access point, without the need for a separate router and Wi-Fi network, to run an independent web server that obeys our commands. To begin, add the latest Pico W MicroPython firmware to the board. Then run the code given below, where you can turn the onboard LEDs on or off using web URLs. Running the code will dump an IP address to access your Pico web server. Just connect to the RPICOWAP Wi-Fi network and execute:

```
curl <ip of pico webserver>/light/on
curl <ip of pico webserver>/light/off
```

…to turn the onboard LED on and off, respectively.

```
import network
import time
import socket
from machine import Pin
from time import sleep

led = Pin("LED", Pin.OUT, value=0)

html = """<!DOCTYPE html>
            <html>
              <head> <title>RPico W</title> </head>
```

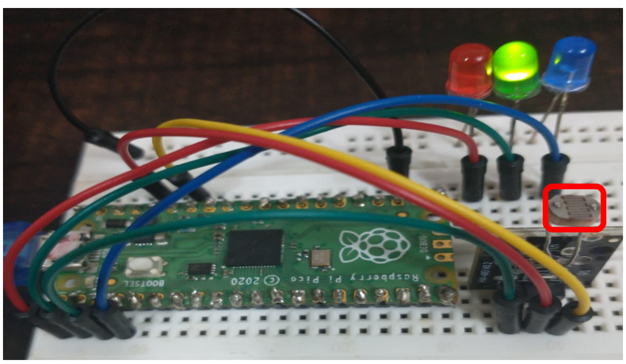

Figure 6: External light sensing

```
<body> <h1>RPico W</h1>
                  <p>%s</p>
                </body>
            </html>
        """

def ap_mode(ssid, password, html, led):
  ap = network.WLAN(network.AP_IF)
  ap.config(essid=ssid, password=password)
  ap.active(True)

  while not ap.active():
    print('Waiting for AP Mode Activation...')
    led.on()
    sleep(0.5)
    led.off()
    sleep(0.5)
  led.off()
  print('AP Mode Is Active, You can Now Connect')
  print('IP Address To Connect to:: ' + ap.ifconfig()[0])

  s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
  s.bind(('', 80))
  s.listen(5)

  while True:
    try:
      conn, addr = s.accept()
      print('Got a connection from %s' % str(addr))
      request = conn.recv(1024)
      print('Content = %s' % str(request))
      response = html

      request = str(request)
      led_on = request.find('/light/on')
      led_off = request.find('/light/off')
      print( 'led on = ' + str(led_on))
      print( 'led off = ' + str(led_off))

      stateis = ''
      if led_on == 6:
        print("led on")
        led.on()
        stateis = "LED is ON"


      if led_off == 6:
        print("led off")
        led.off()
        stateis = "LED is OFF"

      response = html % stateis
```